

Q128.128: Deterministic 256-bit Fixed-Point Arithmetic with Round-to-Nearest-Even Multiplication

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Abstract

We present Q128.128, a deterministic 256-bit signed fixed-point arithmetic engine designed for bit-exact reproducibility across platforms. The representation uses 128 fractional bits, providing ~ 38.5 decimal digits of fractional precision and a sub-Planck resolution of $2^{-128} \approx 2.94 \times 10^{-39}$. Multiplication is performed via exact 512-bit intermediate products followed by round-to-nearest-even (RNE) truncation, achieving an error bound of $|xy - \text{fl}(xy)| \leq 2^{-129}$ (half ULP). The engine includes RamseyIdentity transforms ($E = mc^2 \leftrightarrow i \leftrightarrow E = mc^{-2}$), WASM portability via wasm32-wasi, Vulkan GPU compute via SPIR-V shaders, hardware auto-detection, and a polyglot runtime supporting Python, JavaScript, Rust, C, C#/.NET, and Q#. We verify bit-exact results across native Zig, WASM/wasm3, and Vulkan GPU backends. The engine is deployed as part of the FANO-1 operating system (Puppy Linux/woof-CE fork) with Orange Pi Zero 3W NPU backend support.

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1 Introduction

IEEE-754 floating-point arithmetic [3], while ubiquitous, suffers from non-determinism across platforms due to varying rounding modes, fused multiply-add (FMA) instructions, and compiler optimizations [1, 4]. For applications requiring bit-exact reproducibility—particularly mathematical physics verification—this non-determinism is unacceptable.

Q128.128 addresses this by using integer-only arithmetic with a fixed-point representation. All operations are performed on 256-bit signed integers, with multiplication using exact 512-bit intermediate products and round-to-nearest-even (RNE) truncation. The result is deterministic: the same inputs always produce the same outputs, regardless of platform, compiler, or hardware.

2 Representation

Definition 1 (Q128.128 Format). *A Q128.128 number is a 256-bit two’s-complement signed integer with 128 fractional bits:*

$$x = \frac{\text{raw}(x)}{2^{128}} \quad (1)$$

where $\text{raw}(x) \in [-2^{127}, 2^{127} - 1]$.

Table 1: Q128.128 format parameters.

Parameter	Value
Fractional bits (FRAC_BITS)	128
Scale	$2^{128} \approx 3.40 \times 10^{38}$
Integer range	$\pm 2^{127} \approx \pm 1.70 \times 10^{38}$
Fractional resolution	$2^{-128} \approx 2.94 \times 10^{-39}$
Decimal digits (fractional)	~ 38.5
Internal type	i256 (two’s complement)

The fractional resolution of $2^{-128} \approx 2.94 \times 10^{-39}$ is approximately 10^4 times smaller than the Planck length ($\ell_P \approx 1.62 \times 10^{-35}$ m), providing sub-Planck precision for physics computations [2].

3 Multiplication

3.1 Exact 512-bit Product

Multiplication of two 256-bit integers produces a 512-bit product. The Q128.128 engine computes this exactly using 64-bit-limb schoolbook multiplication:

$$P = a \times b = \sum_{i=0}^3 \sum_{j=0}^3 a_i \cdot b_j \cdot 2^{64(i+j)} \quad (2)$$

where $a = \sum_{i=0}^3 a_i \cdot 2^{64i}$ and $b = \sum_{j=0}^3 b_j \cdot 2^{64j}$ are decomposed into 64-bit limbs.

3.2 Round-to-Nearest-Even Truncation

The 512-bit product P must be truncated to 256 bits by shifting right by 128 (the number of fractional bits). The Q128.128 engine uses round-to-nearest-even (RNE), also known as banker's rounding:

Algorithm 1 RNE Multiplication

Require: $a, b \in \text{Q128.128}$

Ensure: $c = \text{fl}(a \times b)$

```

1:  $P \leftarrow a_{\text{raw}} \times b_{\text{raw}}$                                 ▷ Exact 512-bit product
2:  $c_{\text{trunc}} \leftarrow \lfloor P/2^{128} \rfloor$                         ▷ Truncated result
3:  $d \leftarrow P \bmod 2^{128}$                                     ▷ Discarded low bits
4:  $h \leftarrow 2^{127}$                                           ▷ Halfway point
5: if  $d > h$  then
6:    $c_{\text{raw}} \leftarrow c_{\text{trunc}} + 1$                                 ▷ Round up
7: else if  $d = h$  and  $c_{\text{trunc}}$  is odd then
8:    $c_{\text{raw}} \leftarrow c_{\text{trunc}} + 1$                                 ▷ Round to even
9: else
10:   $c_{\text{raw}} \leftarrow c_{\text{trunc}}$                                     ▷ Truncate
11: end if
12: return  $c$ 

```

Theorem 1 (Error Bound). *The RNE multiplication satisfies:*

$$|xy - \text{fl}(xy)| \leq 2^{-129} \quad (3)$$

i.e., at most half a unit in the last place (ULP).

Proof. The discarded bits $d \in [0, 2^{128} - 1]$. The halfway point is $h = 2^{127}$. RNE ensures $|d - h| \leq h$ after rounding, so the error is at most $h/2^{128} = 2^{127}/2^{128} = 2^{-1}$ ULP = 2^{-129} in value. $\square$ $\square$

3.3 Overflow Handling

If the product exceeds the representable range, the result saturates to Q128_MAX or Q128_MIN with an overflow flag set.

4 Division and fromRatio

Division is truncating (floor toward zero):

$$\text{fl}(a/b) = \left\lfloor \frac{a_{\text{raw}} \times 2^{128}}{b_{\text{raw}}} \right\rfloor \quad (4)$$

The `fromRatio` constructor creates a Q128.128 from a rational n/d using round-half-away-from-zero:

The condition $r \neq 0$ in the round-half-away-from-zero check prevents an incorrect round-up when $n = 0$ and $d = 1$ (where both r and h are zero). This was identified and fixed during the archive audit.

Algorithm 2 fromRatio

Require: $n, d \in \mathbb{Z}, d \neq 0$ **Ensure:** $x = \text{fl}(n/d)$ in Q128.1281: $\text{sign} \leftarrow \text{sgn}(n) \cdot \text{sgn}(d)$ 2: $q \leftarrow \lfloor |n| \times 2^{128} / |d| \rfloor$ 3: $r \leftarrow (|n| \times 2^{128}) \bmod |d|$ 4: $h \leftarrow |d|/2$ $\triangleright$ Integer division5: **if** $r > h$ **or** ($r = h$ **and** $r \neq 0$) **then**6: $q \leftarrow q + 1$ $\triangleright$ Round half away from zero7: **end if**8: **return** $\text{sign} \times q$

5 Square Root

Square root is computed via Newton’s method iteration:

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{S}{x_n} \right) \quad (5)$$

starting from an initial estimate and converging to $\sqrt{S}$ within a tolerance of 2^{68} (conservation check tolerance).

6 RamseyIdentity Transforms

The Q128.128 engine includes RamseyIdentity transforms that encode the framework’s structural identity:

$$E = mc^2 \leftrightarrow i \leftrightarrow E = mc^{-2} \quad (6)$$

$$i = e_0[e_1, e_2, e_3, e_4, e_5, e_6, e_7] \rightarrow e_0 \quad (7)$$

These transforms map between energy-momentum space ($E = mc^2$), the imaginary unit (i), and the inverse energy relation ($E = mc^{-2}$), with the octonionic identity providing the algebraic bridge.

7 WASM Portability

The Q128.128 engine targets wasm32-wasi for cross-platform browser and WASM runtime verification:

Table 2: WASM target specifications.

Parameter	Value
Target	wasm32-wasi
Module size	598 KB
Exported functions	10 (integer-only proof checks)
Runtime	wasm3, custom Zig interpreter
Verification	Bit-exact with native

The full 30-module proof suite uses i256 arithmetic, which requires compiler-rt library calls not available in wasm32-wasi with Zig 0.13. The WASM module exports 10 integer-only proof checks that run in any WASM runtime. For full verification, the native binary is recommended.

8 Vulkan GPU Compute

The Q128.128 engine includes Vulkan compute shaders (SPIR-V) for GPU-accelerated fixed-point multiplication:

Table 3: Vulkan GPU compute verification.

Parameter	Value
GPU	AMD Radeon RX 580
Shader	SPIR-V compute
Nodes verified	4096
Result	Bit-exact with CPU
Backend	Vulkan 1.3 (also PowerVR BXM-4-64)

The GPU pipeline uploads Q128.128 values as storage buffers, dispatches a compute shader that performs 64-bit-limb schoolbook multiplication and RNE truncation, and reads back results for comparison with the CPU implementation.

9 Hardware Auto-Detection

The FANO-1 runtime includes hardware auto-detection (`fano-detect`) that queries:

- CPU features (AVX, SSE, NEON)
- Vulkan devices and VRAM
- NPU presence (VeriSilicon VIP9000 on Orange Pi Zero 3W)
- Available memory

Based on detection results, the runtime selects the optimal backend:

1. NPU (VIP9000, 3 TOPS @ INT8)—for LLM inference
2. Vulkan GPU—for compute shaders
3. Native Zig (CPU)—for full proof suite
4. WASM (wasm3)—for fallback and browser

10 Polyglot Runtime

The polyglot runtime allows Q128.128 functions to be called from multiple programming languages in-process:

Table 4: Polyglot runtime support.

Language	Integration	Tests
Python	CPython C API	22/22
JavaScript	QuickJS engine	22/22
Rust	FFI	22/22
C	Standard C ABI	22/22
C#/.NET	P/Invoke	22/22
Q#	Subprocess bridge	51 witnesses

11 FANO-1 OS Deployment

The Q128.128 engine is deployed as part of FANO-1, a Puppy Linux (woof-CE) fork designed for deterministic computing:

Table 5: FANO-1 OS deployment targets.

Platform	Hardware	Backend
Desktop	x86_64 + RX 580	Vulkan GPU
Orange Pi Zero 3W	Allwinner A733	NPU + Vulkan
Browser	Any	WASM (wasm3)

The Orange Pi Zero 3W includes a VeriSilicon VIP9000 NPU (3 TOPS @ INT8) with INT8/INT16/FP16/BF16 mixed-precision support, enabling on-device LLM inference (SmolLM2-135M at 21 tokens/second).

12 Benchmark Results

Table 6: Test results summary.

Component	Tests	Status
Q128.128 engine	28	ALL PASS
Physics modules	393	ALL PASS
Polyglot runtime	22	ALL PASS
FANO-1 hooks	37	ALL PASS
WASM module	10	ALL PASS
Q# witnesses	51	ALL PASS
Total	541	ALL PASS

13 Comparison with IEEE-754

14 Conclusion

Q128.128 provides deterministic 256-bit fixed-point arithmetic with round-to-nearest-even multiplication, achieving bit-exact reproducibility across native, WASM, and GPU platforms. The engine’s

Table 7: Comparison: IEEE-754 double vs Q128.128.

Property	IEEE-754 double	Q128.128
Total bits	64	256
Fractional bits	52	128
Decimal digits	~ 15.9	~ 38.5
Integer range	$\pm 2^{1023}$	$\pm 2^{127}$
Fractional resolution	$2^{-52} \approx 2.2 \times 10^{-16}$	$2^{-128} \approx 2.9 \times 10^{-39}$
Determinism	Platform-dependent	Bit-exact
Reproducibility	Not guaranteed	Guaranteed
Rounding	Implementation-defined	RNE (specified)
Overflow	$\pm\infty$	Saturation

2^{-129} error bound, sub-Planck resolution, and polyglot runtime make it suitable for mathematical physics verification where IEEE-754 floating-point non-determinism is unacceptable. The FANO-1 OS deployment with Orange Pi Zero 3W NPU support enables edge deployment of deterministic computing workloads.

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